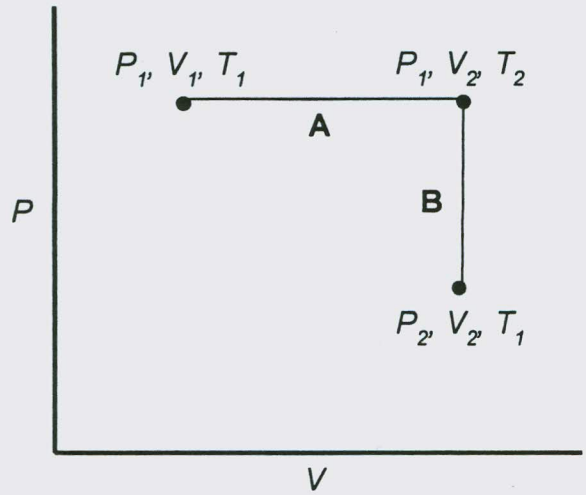
Physical Chemistry

Thermodynamics

February 14, 2014

1. (50 pts) Calculate q_{rev} and ΔS for a reversible expansion of an ideal gas at constant pressure P_1 to T_1, V_1 to T_2, V_2 (path A) followed by a reversible cooling of the gas at constant volume V_2 from P_1, T_2 to P_2, T_1 (path B).



2. (50 pts) At 0 °C, ice absorbs 333.5 J g^{-1} in melting and water absorbs 2490 J g^{-1} in vaporizing.
   (a) What is the heat of sublimation of ice at this temperature?
   (b) At 0 °C, the vapor pressure of both ice and water is 611 Pa. What is the rate of change of vapor pressure with temperature dP/dT for ice and liquid water at this temperature?
   (c) Estimate the vapor pressures of ice and of liquid water at -5 °C.

(Hint: You can start with Clapeyron equation $\frac{dp}{dT} = \frac{\Delta_{trs} S}{\Delta_{trs} V}$ to solve the problem.)